

Qihao Qian

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EDUCATION

University of California, San Diego (UCSD)

Master of Science in Electrical Computer Engineering, Machine Learning and Data Science

Sep 2024 – June 2026

San Diego, US

University of Electronic Science and Technology of China (UESTC)

Bachelor of Engineering in Software Engineering, Major GPA: 3.77

Sep 2020 – June 2024

Chengdu, China

PUBLICATION

- Z. Dai*, **Q. Qian***, T. Fan, N. Atanasov, “ ∇ -SDF: Learning Euclidean Signed Distance Functions Online with Gradient-Augmented Octree Interpolation and Neural Residual.” Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. *Equal contribution. (ArXiv: 2510.18999)
- **Q. Qian**, “RailYolact: A YOLACT Focused on Edge for Real-Time Rail Segmentation.” (ArXiv: 2410.09612).

RESEARCH EXPERIENCE

∇ -SDF: Online Euclidean SDF Mapping

Jan 2025 – Present

Existential Robotics Lab (UC San Diego) | Advisor: Prof. Nikolay Atanasov

San Diego, US

- Designed a **semi-sparse octree** that improves memory efficiency and interpolation accuracy (**-45%** SDF MAE, **-38%** Grad. MAE) compared with the sparse octree, enabling faster training and smoother surface reconstruction.
- Proposed a **projection loss** that provides supervision for free-space regions, improving reconstruction accuracy by over **6×** in Chamfer distance and **+14%** in F1 score on the Replica dataset while achieving faster convergence.
- Implemented all the training and evaluation pipeline and conducted extensive ablation and baseline experiments, demonstrates that **∇ -SDF** outperforms the state of the art in terms of accuracy and efficiency.
- Validated the model on the challenging Newer College dataset and successfully **deployed the pipeline on a Clearpath Jackal robot** and in simulation, demonstrating robust real-world performance in complex environments.
- Integrated the SDF-based mapping with an A* planning on a quadrotor, leveraging the distance field to achieve faster collision checking and smoother trajectory generation in constrained environments.

3D Gaussian Splatting for 6D Pose Estimation

Mar 2024 – Aug 2024

Peking University, School of Intelligence Science and Technology

Beijing, China

- Implemented **3D Gaussian Splatting (3DGS)** within the FoundationPose framework, replacing the standard SDF-based Neural Object Modeling to test hypothesis on appearance-based optimization.
- Conducted comparative analysis revealing that while 3DGS improved texture rendering, the lack of precise depth supervision caused a degradation in geometric fidelity, negatively impacting pose estimation.
- Redirected research focus to **Sparse View 3DGS** for modeling objects from extremely sparse inputs (1-2 views), aiming to optimize the efficiency of the model-free pose estimation pipeline.

Real-time Rail Segmentation Focus On Edge

Mar 2023 – Oct 2023

UESTC, School of Mathematical Sciences

Chengdu, China

- Developed **RailYolact**, an edge-aware real-time rail segmentation algorithm for intelligent rail obstacle warning.
- Enhanced the YOLACT model by integrating Laplacian-based **edge supervision** and a box-filter **mask smoothing** strategy, improve accuracy on the Cityscapes dataset (**+4.1 AP**, **+4.6 AP_{50}**) and our Railway Dataset (**+1.0 AP**) while maintain same speed as YOLACT (28FPS).

Object Detection for Military Aircraft Remote Sensing Images

Oct 2023 – Feb. 2024

UESTC, School of Information and Software Engineering

Chengdu, China

- Improve target detection accuracy by replacing the SPPF module in YOLOv8 with a proposed SPPFCSPC module and introducing the Double Attention module before the detection head for military aircraft remote sensing image target detection tasks, resulting in a **2.6%** increase in mAP.

INDUSTRY EXPERIENCE

XbrainFeb. 2024 – Mar. 2024

AI Engineer InternNanjing, China

- Generate realistic, lip-synchronized lower-face video from speech audio
- Processed and segmented over **1,000 hours** of video data using **RetinaNet** for face detection and segmentation, and aligned corresponding audio with detected faces to construct paired training data.
- Replaced the Mel-spectrogram with the pre-trained **HuBERT** model for high-level audio feature extraction and upgraded the baseline U-Net to the **U-Mamba** framework, achieving a **6.7%** reduction in generation loss for realistic lower-face synthesis.

China Institute of People’s Life SafetyFeb. 2023 – June. 2023

AI Engineer InternChengdu, China

- Utilized the **LLaMA** model to automatically structure unstructured medical paper abstracts into four sections: background, methods, subjects, and conclusions, supporting subsequent analysis tasks.
- Generated 5000 training samples using the **GPT-3.5 API** to enhance LLaMA’s adaptability to the task.
- Employed **LoRA** (Low-Rank Adaptation) for efficient training, optimizing computational resource usage.

DJIJune 2022 – Dec 2022

AI Engineer InternShenzhen, China

- Contributed to a web-based **lane marking annotation system** that allows users to intuitively label lane markings on 3D point cloud data; the annotated coordinates are automatically stored in a database for downstream task.
- Implemented and trained a **PointNet++** model on the Toronto3D dataset to segment lane line from point clouds, using data augmentation (rotation, translation, scaling, and perturbation) to improve generalization and achieving **78.5%** overall accuracy on the test set.
- Integrated the trained model into the backend via **FastAPI**, enabling real-time inference from the web interface.

HONORS & AWARDS

Outstanding Student Scholarship, UESTC2021 & 2023

University Arts Group Scholarship, UESTC2022

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, etc.
AI Models: CLIP, LLAMA3, DINO, GroundedSAM, YOLO, 3D Guassian, NeRF,etc.
Developer Tools: Linux, Docker, Git, Kubernetes, Cursor, etc.
Libraries: PyTorch, ROS2, OpenCV, Open3D, Gazebo, Transformers, Matplotlib, etc.